

Where the project is now

The project has narrowed from a broad artificial-society idea to a focused social-ecological experiment.

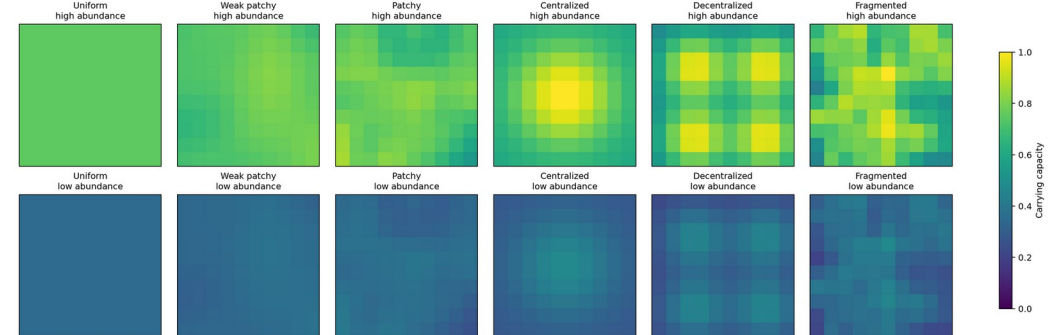
Current core

Adaptive social information in a renewable common-pool resource world

Agents learn how much to extract, and some treatments let them change who they observe.



Capacity-pattern families



Ecological variation remains the testbed: abundance and spatial structure define the resource problem.

Distilled research question

How does prediction-error-driven rewiring of who agents observe affect resource sustainability, welfare, and inequality?

Compared against fixed networks and matched random turnover.

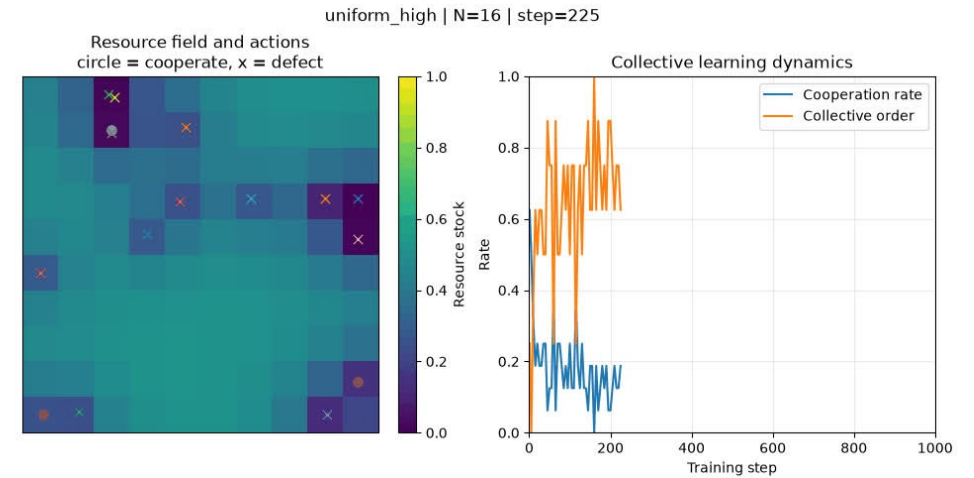


Secondary question: how do the effects change across ecological conditions and local vs. global search?

Current model structure

The model separates the physical resource world from the social information layer.

- 1 Resource world**
Renewable grid with local stock, carrying capacity, regeneration, depletion, and coupling.
- 2 Agents**
Independent tabular Q-learners choose low or high extraction.
- 3 Social layer**
Agents observe previous actions from a limited set of sources.
- 4 Network update**
Prediction error can trigger source replacement.



Treatment design

The design compares no-social learning, fixed social information, random turnover, and adaptive rewiring.

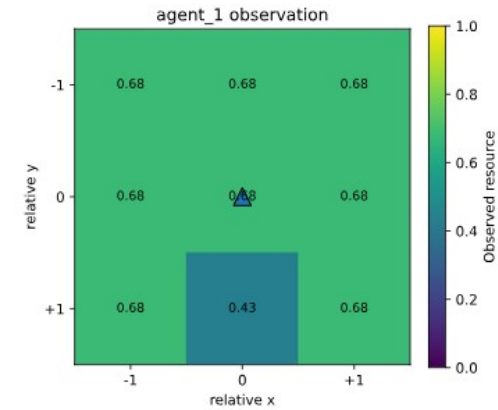
ID	Social information	Network	Dynamics
B0	None	None	Independent ecological learning
S1	Peer actions	Random fixed-k	Fixed observation
S2	Peer actions	BA-style	Fixed visibility-skew control
R0	Peer actions	Random fixed-k	Matched random turnover
R1–R3	Peer actions	Random fixed-k	Prediction-error rewiring; local to global search

Key comparison: adaptive rewiring vs. event-count-matched random rewiring, under the same ecological and learning settings.

Implementation update

Core mechanisms are now in place.

- **Ecology** Renewable resource dynamics and spatial scenario families
- **Learning** Independent tabular Q-learning with low/high extraction
- **Social signals** Previous peer actions become a discretized social state
- **Rewiring** Prediction error drives local/global source replacement
- **Analysis** Cross-treatment summaries, pairing, and diagnostics



Social observation is intentionally local and limited: agents see only selected sources, not the full population.

Codebase cleanup target

The scientific core is smaller than the repository currently appears.

1,824

tracked files under results/

~135 MB

generated result artifacts

3,606

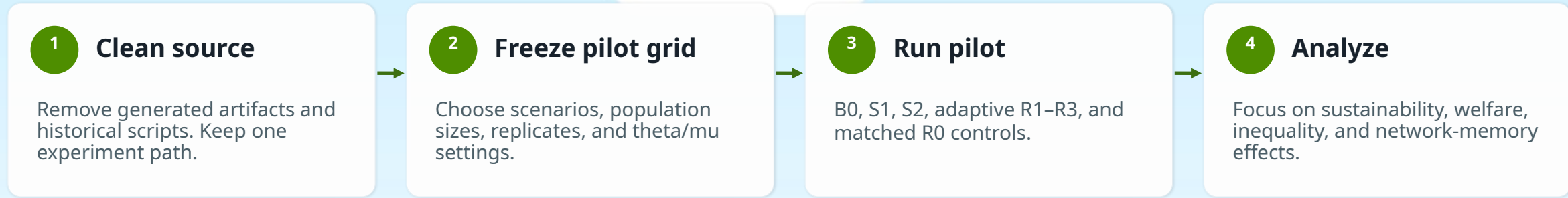
lines in legacy qlearning_experiment.py

Cleanup plan

Remove committed outputs · extract scenarios · delete old scripts · rename package to lowercase · shorten README and move detail to docs

Next sprint

Goal: move from implemented mechanisms to a minimal repo, pilot runs, and a clean analysis story.



Decision to make next: freeze the exact pilot campaign before running long experiments.